# Evaluation runs: **naive** (s1-naive)

generated 2026-11-05 from brain-data bb4df53ad 2026-11-05 15:31:21 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): naive: expected violation; self: expected conforms. Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | conforms | 0.0 | 4 | 1 h, 250 GB | 2 | 3 | brain03, 2026-11-02 |
| 3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 4 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 24 | 300 | brain03, 2026-11-02 |
| 5 | 120 | 120 | conforms | 0.3 | 4 | 1 h, 250 GB | 120 | 7260 | brain03, 2026-11-02 |
| 6 | 720 | 720 | conforms | 9.4 | 4 | 1 h, 250 GB | 720 | 259560 | brain03, 2026-11-02 |
| 7 | 5040 | 5040 | conforms | 497.8 | 11 | 1 h, 250 GB | 5040 | 12703320 | brain03, 2026-11-02 |

### stateless: pair **naive**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | 2 | violation | 0.0 | 4 | 1 h, 250 GB | 2 | 4 | brain03, 2026-11-02 |
| 3 | 6 | 6 | 6 | violation | 0.0 | 4 | 1 h, 250 GB | 6 | 36 | brain03, 2026-11-02 |
| 4 | 24 | 24 | 24 | violation | 0.0 | 4 | 1 h, 250 GB | 24 | 576 | brain03, 2026-11-02 |
| 5 | 120 | 120 | 120 | violation | 0.5 | 2 | 1 h, 250 GB | 120 | 14400 | brain03, 2026-11-02 |
| 6 | 720 | 720 | 720 | violation | 18.4 | 9 | 1 h, 250 GB | 720 | 518400 | brain03, 2026-11-02 |
| 7 | 5040 | 5040 | 5040 | violation | 981.9 | 26 | 1 h, 250 GB | 5040 | 25401600 | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): naive 8 1 bag, self 8 1 bag, naive 8 2 bag, self 8 2 bag, naive 9 1 bag, self 9 1 bag, self 9 2 bag, naive 9 2 bag, self 10 1 bag, naive 10 1 bag, naive 10 2 bag, self 10 2 bag.

left out by the table rules (ok): naive 2 2 bag, self 2 2 bag, naive 3 2 bag, self 3 2 bag, naive 4 2 bag, self 4 2 bag, self 5 2 bag, naive 5 2 bag, self 6 2 bag, naive 6 2 bag, naive 7 2 bag, self 7 2 bag.

left out by the table rules (invalid: foreign load): naive 10 bag, naive 11 bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 24 | 24 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 120 | 120 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 720 | 720 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 7 | 5040 | 5040 | conforms | 0.4 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **naive**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | 2 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 24 | 24 | 24 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 120 | 120 | 120 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 720 | 720 | 720 | violation | 0.1 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7 | 5040 | 5040 | 5040 | violation | 0.6 | 34 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8 | 40320 | 40320 | 40320 | violation | 5.1 | 267 | 1 h, 250 GB | brain02, 2026-11-02 |
| 9 | 362880 | 362880 | 362880 | violation | 53.7 | 3211 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10 | 3628800 | 3628800 | 3628800 | violation | 609.5 | 41493 | 1 h, 250 GB | brain02, 2026-11-02 |
| 10 | 3628800 | 3628800 | 3628800 | violation | 664.1 | 41493 | 48 h, 250 GB | brain04, 2026-11-03 |
| 11 | | | | oom (250 GB) | | | 48 h, 250 GB | brain01, 2026-11-03 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): naive 8 1 bag, self 8 1 bag, naive 8 2 bag, self 8 2 bag, naive 9 1 bag, self 9 1 bag, self 9 2 bag, naive 9 2 bag, naive 10 1 bag, self 10 1 bag, naive 10 2 bag, self 10 2 bag, self 14 2 bag, naive 14 2 bag.

left out by the table rules (ok): naive 2 2 bag, self 2 2 bag, naive 3 2 bag, self 3 2 bag, naive 4 2 bag, self 4 2 bag, self 5 2 bag, naive 5 2 bag, self 6 2 bag, naive 6 2 bag, naive 7 2 bag, self 7 2 bag, naive 8 2 bag, naive 9 2 bag, naive 10 2 bag.

the tool ended with an error (exit-101): self 8 1 bag.

skipped: a smaller size had reached its limit: self 8 2 bag, self 9 1 bag, self 9 2 bag.

## offline

### offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 6 | 6 | conforms | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 24 | 24 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5 | 120 | 120 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 6 | 720 | 720 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 7 | 5040 | 5040 | conforms | 0.6 | 10 | 1 h, 250 GB | 0.3 | 0.4 | 0.3 | merge | brain01, 2026-11-03 |

### offline: pair **naive**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | 2 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2 |
| 3 | 6 | 6 | 6 | violation | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain01, 2 |
| 4 | 24 | 24 | 24 | violation | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain01, 2 |
| 5 | 120 | 120 | 120 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2 |
| 6 | 720 | 720 | 720 | violation | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2 |
| 7 | 5040 | 5040 | 5040 | violation | 0.7 | 60 | 10 min, 16 GB | 0.3 | 0.4 | 0.3 | in-memory | brain01, 2 |
| 10 | 3628800 | 3628800 | 3628800 | violation | 730.9 | 9811 | 48 h, 250 GB | 384.9 | 397.6 | 333.3 | merge | brain01, 2 |
| 11 | 39916800 | 39916800 | 39916800 | violation | 13349.0 | 115422 | 48 h, 250 GB | 5218.0 | 5447.0 | 7902.0 | merge | brain01, 2 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): naive 8 1 bag, self 8 1 bag, naive 8 2 bag, self 8 2 bag, naive 9 1 bag, self 9 1 bag, self 9 2 bag, naive 9 2 bag, self 10 1 bag, naive 10 1 bag, naive 10 2 bag, self 10 2 bag.

left out by the table rules (ok): naive 2 2 bag, self 2 2 bag, naive 3 2 bag, self 3 2 bag, naive 4 2 bag, self 4 2 bag, self 5 2 bag, naive 5 2 bag, self 6 2 bag, naive 6 2 bag, naive 7 2 bag, self 7 2 bag.